fourth degree and cannot be treated analytically in a simple way, but may be solved in each instance by approximate or numerical methods. In addition, there is an ambiguity in that there are two real roots rather than one. This can be seen from Figure 9.14, where the Hall coefficient is plotted as a function of p_0; for a given value of R there are always *two* possible values of p_0. In practice, however, it is usually fairly easy to choose which root is the right one by making a series of Hall effect

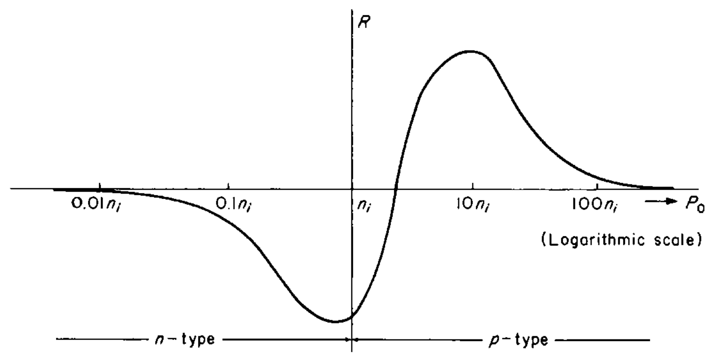

FIGURE 9.14. Hall coefficient of a semiconductor crystal as a function of hole concentration.

measurements over a wide range of temperatures. In addition, in order to obtain the carrier concentrations by the above procedure, it is necessary to know the mobility ratio b. This may be ascertained by making separate measurements of Hall mobility in highly extrinsic n- and p-type samples, or from the results of excess carrier drift measurements, which will be discussed later. If n_0 and p_0 are ascertained from the Hall coefficient in this way, and if the ratio $\mu_n/\mu_p = b$ is known, the mobilities μ_n and μ_p may be determined from conductivity data, by means of (9.7-8).

9·9 CYCLOTRON RESONANCE AND ELLIPSOIDAL ENERGY SURFACES

The electrons and holes in semiconductors may be made to exhibit a resonance in the presence of a large constant magnetic field (which we shall assume to lie along the z-axis) and a small transverse oscillating radiofrequency or microwave electromagnetic field whose electric vector lies in the xy-plane. The resulting hole or electron orbits are circular at resonance, and energy is absorbed from the rf-field every half cycle, as in a cyclotron. The situation is, in fact, precisely analogous to what happens to nuclear charged particles in a cyclotron, and for this reason the phenomenon is termed *cyclotron resonance.*

Let us initially investigate the case of spherical energy surfaces and a single scalar effective mass. The Lorentz force on a hole (and we shall consider holes in a p-type sample in this specific calculation) is given by (9.8-4). If the effective mass m_p^* is independent of direction, and if the electric vector of the oscillating field, whose frequency is ω, is assumed to vibrate along the x-direction, then

$$E_x = E_0 e^{i\omega t}; \qquad E_y = E_z = 0 \tag{9.9-1}$$

while

$$B_z = B_0; \qquad B_x = B_y = 0, \tag{9.9-2}$$

and the components of the Lorentz force give for the equations of motion

$$F_x = m_p^* \frac{d^2x}{dt^2} = \frac{e}{c} v_y B_0 + eE_0 e^{i\omega t} \tag{9.9-3}$$

$$F_y = m_p^* \frac{d^2y}{dt^2} = -\frac{e}{c} v_x B_0 \tag{9.9-4}$$

$$F_z = m_p^* \frac{d^2z}{dt^2} = 0. \tag{9.9-5}$$

In writing these equations it has been assumed that the magnetic vector of the rf-field is negligible compared with the constant field $\mathbf{B}_0$. The z-component equation (9.9-5) tells us merely that the particle moves with constant velocity along the z-direction; this is of no particular interest and we need trouble ourselves no further about it. The x- and y-component equations may be written

$$\frac{d^2x}{dt^2} = \omega_0 \frac{dy}{dt} + \frac{eE_0}{m_p^*} e^{i\omega t} \tag{9.9-6}$$

$$\frac{d^2y}{dt^2} = -\omega_0 \frac{dx}{dt} \tag{9.9-7}$$

where ω_0 is the very same frequency defined by (9.8-7) in connection with the Hall effect.

If oscillatory solutions of the form $x = x_0 e^{i\omega t}$ and $y = y_0 e^{i\omega t}$ are assumed these equations reduce to

$$i\omega\omega_0 y_0 + \omega^2 x_0 = -eE_0/m_p^* \tag{9.9-8}$$

and

$$-\omega^2 y_0 + i\omega\omega_0 x_0 = 0 \tag{9.9-9}$$

which can be solved for the amplitudes x_0 and y_0 to give

$$x_0 = \frac{eE_0/m_p^*}{\omega_0^2 - \omega^2} \tag{9.9-10}$$

$$y_0 = \frac{i\omega_0}{\omega} x_0 = \frac{i\omega_0 eE_0/m_p^*}{\omega(\omega_0^2 - \omega^2)}. \tag{9.9-11}$$

When $\omega = \omega_0 = eB_0/m_p^* c$ resonance occurs and the amplitudes become very large.

The resonance frequency ω_0 is often referred to as the *cyclotron frequency*. Clearly, if ω_0 is measured experimentally and if the magnetic induction B_0 is accurately known, the effective mass m_p^* can be determined. In practice, the cyclotron resonance effect affords one of the best methods of measuring effective masses. From (9.9-10) and (9.9-11), it can be seen that as $\omega \to \omega_0$

$$y_0 \cong ix_0 = x_0 e^{i\pi/2} \tag{9.9-12}$$

whereby

$$x(t) = x_0 e^{i\omega_0 t} \tag{9.9-13}$$

$$y(t) = y_0 e^{i\omega_0 t} = x_0 e^{i(\omega_0 t + \pi/2)}. \tag{9.9-14}$$

At resonance, then, $x(t)$ and $y(t)$ are orthogonal harmonic vibrations of equal amplitude which differ in phase by 90°. It is easily seen that the resultant particle trajectory, representing the hole orbit, is circular.

In practice, the resonance is usually observed by measuring the Q-factor of a microwave resonant cavity as the magnetic field B_0 (and hence the frequency ω_0) is varied. When ω_0 equals the microwave excitation frequency, a sharp decrease in the Q of the cavity is obtained, because at resonance a great deal of electromagnetic field energy is absorbed when carriers are excited to large resonance amplitudes and then are scattered by impurities or lattice vibrations. It is worth noting that in this experiment one wishes to make the quantity $\omega_0\tau$ much *larger* than unity, so that the carriers can be excited through several complete orbits before being scattered. Under these conditions they are capable of absorbing a great deal of microwave energy and transforming it, when scattering takes place, into vibrations of the crystal lattice. The experiment is usually performed at low temperature, so as to obtain a large value of τ, and with as large a static field B_0 as is possible consistent with the requirement that ω_0 be an experimentally accessible microwave frequency. In the case of Hall effect measurements, on the other hand, the objective is often to make $\omega_0\tau$ much *less* than unity so as to simplify the interpretation of the data.

When the cyclotron resonance effect in silicon and germanium was studied experimentally, it was found for both substances that there were *more* than two resonances (one for electrons, one for holes) as predicted by the simple theory discussed above.[5,6] Moreover, it was observed that some of the cyclotron resonance frequencies underwent significant changes as the orientation of the crystal with respect to the static magnetic field was altered, while others were practically independent of sample orientation. These effects were finally explained using a model in which the constant energy surfaces for electrons in **k**-space are ellipsoidal rather than spherical, and in which the presence of three separate valence bands, two of which are degenerate at $k = 0$, are considered. The constant energy surfaces for holes in these valence bands are roughly spherical, although since $\partial^2\varepsilon/\partial k^2$ differs for the different valence bands there is more than one possible effective mass for holes. The various features of this model have been verified not only by the agreement with experimental data which has been obtained, but also by quantum-mechanical calculations of the energy band structure which begin, more or less, from first principles.[7,8,9]

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[5] G. Dresselhaus, A. F. Kip and C. Kittel, *Phys. Rev.*, **92**, 827 (1953).
[6] R. N. Dexter, H. J. Zeiger and B. Lax, *Phys. Rev.*, **104**, 637 (1956).
[7] F. Herman, *Phys. Rev.*, **93**, 1214 (1954); *Proc. Inst. Radio Engrs.*, **43**, 1703 (1955).
[8] D. P. Jenkins, *Physica*, **20**, 967 (1954).
[9] E. M. Conwell, *Proc. Inst. Radio Engrs.*, **46**, 1281 (1958).

We shall first consider the matter of ellipsoidal energy surfaces, and investigate how the results of the cyclotron resonance experiment are modified when the energy surfaces have this form. Up to this point we have always assumed that the minimum energy point in the reduced Brillouin zone occurs at $k = 0$, in the center of the zone, as illustrated in Figure 8.6. In this case an electron near the energy minimum exhibits the dynamical behavior of a free electron, leading to constant energy surfaces which, within this region of **k**-space, are approximately spherical, as shown in Figure 8.11(a).

It has been found, however, that the minimum value of energy within the reduced zone need not necessarily be at $k = 0$, but may lie elsewhere within the zone. This is indeed the observed state of affairs for electrons in the conduction band of germanium and silicon. Since the diamond lattice is really two interpenetrating face-centered cubic lattices separated along the cube diagonal by a distance $a\sqrt{3}/4$ (where a is the cube edge), the primitive unit cell for this lattice is the same as that shown in Figure 1.5, except that there is a second atom within the cell. The reciprocal lattice is thus the same as that for the f.c.c. structure, and hence the Brillouin zone has the same shape as the f.c.c. zone shown in Figure 8.14(c). A plot of ε *versus* k along the (100) direction in this Brillouin zone for electrons in the conduction band of silicon is shown in Figure 9.15(a). The smallest value of ε is reached at a value of k which is about

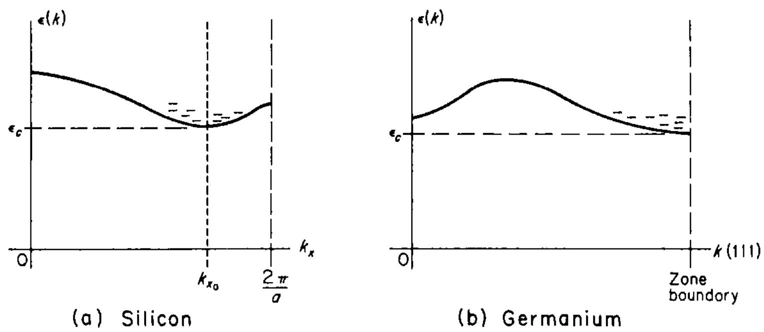


FIGURE 9.15. The ε *versus* k relation for (a) the conduction band of silicon, plotted along the k_x-direction, (b) the conduction band of germanium, plotted along a (111) direction in **k**-space.

$0.8(2\pi/a)$. This is the absolute minimum value of ε within the Brillouin zone; curves of ε *versus* k along other paths may exhibit minimum values of ε, but none so low as the value ε_c in the figure. For energies greater than ε_c, one may construct constant energy surfaces about the minimum energy point. It is clear that the energy *versus* crystal momentum curve shown in Figure 9.15(a) is roughly parabolic about the minimum point, corresponding to free electron behavior with an appropriate effective mass which we shall call $m_{\parallel}^*$. If one were to plot the ε *versus* k curve along some path passing through the minimum point, but *perpendicular* to the k_x axis (for example a line parallel to the k_y axis) one would also obtain a curve which would be parabolic about the minimum point, but there is now no aspect of crystal symmetry which would require that the *curvature* in this direction be the same as it is along the k_x direction. (This situation is illustrated in Figure 9.16.) As a result the effective mass related to changes in the momentum component along this direction has a value which is different from the value discussed previously, and which we shall denote as $m_\perp^*$. It turns out, as we shall soon see, that requirements of crystal symmetry demand that the effective mass have the same value $m_\perp^*$ for *any* direction in **k**-space normal to the k_x-direction for this example.

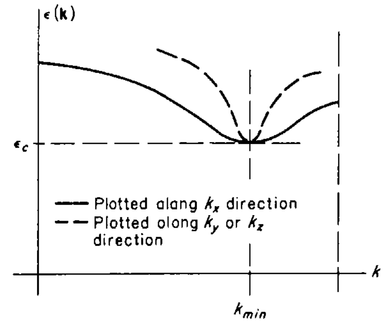


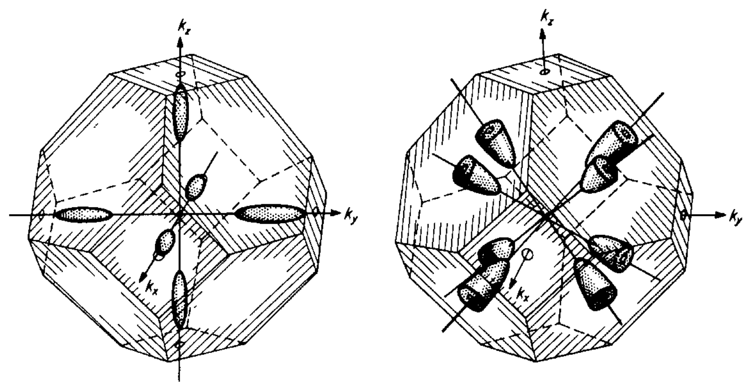
FIGURE 9.16. The ε *versus* k relation for the conduction band of silicon plotted along the k_x-direction (solid curve) and along a line normal to the k_x-axis which passes through the point k_{min} (dashed curve). The different curvatures at the minimum point reflect the anisotropy of the effective mass.

The energy in excess of ε_c represents the kinetic energy of a free electron in the conduction band, whereby

$$\varepsilon - \varepsilon_c = \frac{(p_x - p_{x0})^2}{2m_\parallel^*} + \frac{(p_y^2 + p_z^2)}{2m_\perp^*} \tag{9.9-15}$$

where $p_{x0} = \hbar k_{x0}$ represents the value of p_x at the bottom of the band, as shown in Figure 9.15(a). According to this the surface of constant energy for an electron of energy ε slightly greater than ε_c is seen to be an *ellipsoid of revolution* whose center is at $k_x = k_{x0}$, $k_y = k_z = 0$. Since, however, the six $\langle 100 \rangle$ directions of a cubic lattice are crystallographically equivalent, the ε *versus* k curve along any of these six directions must be the same. There must then be *six* equivalent energy minima along the six $\langle 100 \rangle$ directions, and associated with each of these minima there must be a family of ellipsoidal constant energy surfaces similar to those described by (9.9-15). This situation is illustrated in Figure 9.17(a). It is now clear why the ellipsoids have to be ellipsoids of revolution; as we saw in the previous chapter, the constant energy surfaces must have all the symmetry properties of the Brillouin zone, which in turn has all the symmetry properties of a cubic crystal. If the ellipsoids were not ellipsoids of revolution, the family of constant energy surfaces would no longer be invariant under all the operations for which a cubic crystal remains invariant.

The situation for electrons in the conduction band of germanium is somewhat similar, except that there are a set of eight equivalent minimum energy points which lie along the $\langle 111 \rangle$ directions in **k**-space at the intersection of those directions with the surface of the Brillouin zone. These minima thus lie at the centers of the hexagonal faces of the Brillouin zone. The constant energy surfaces are again ellipsoidal in form, but since the energy minima are at the zone boundary, and since there is a rather large forbidden energy region which must be surmounted before electrons can be excited outside the Brillouin zone in these directions, there are eight half-ellipsoids which extend into the interior of the Brillouin zone along the $\langle 111 \rangle$ directions, as shown in Figure 9.17(b). For many purposes these may be considered to be equivalent to four full ellipsoids. In both cases there is an effective mass *tensor* associated with *each*



(a) Silicon (b) Germanium

FIGURE 9.17. Ellipsoidal constant energy surfaces in **k**-space for (a) silicon and (b) germanium. In silicon the major axes of the ellipsoids are along {100} directions, while in germanium, since the energy minimum is on the zone boundary, the constant energy surfaces form eight half-ellipsoids whose major axes lie along {111} directions.

ellipsoid whose elements are described by (8.8-7). In calculating the transport properties and other physical characteristics of crystals in which the energy surfaces are ellipsoidal, one usually calculates the effect due to one ellipsoid (which is generally anisotropic) and then sums over all ellipsoids. In cubic crystals, the transport properties are found to be isotropic after the summation is carried out, despite the fact that anisotropies are associated with individual ellipsoids. In both germanium and silicon the longitudinal effect mass $m_\parallel^*$ is much larger than the transverse mass $m_\perp^*$, the ellipsoids thus being quite long and thin. The actual values, as determined by cyclotron resonance experiments,[5] are given in Table 9.2.

TABLE 9.2.

Effective Masses of Electrons in Germanium and Silicon

	$m_\parallel^*$	$m_\perp^*$	"mass ratio" $m_\parallel^*/m_\perp^*$
Germanium	$1.64\ m_0$	$0.0819\ m_0$	20.0
Silicon	$0.98\ m_0$	$0.19\ m_0$	5.2

Let us now investigate the cyclotron resonance effect in materials where the energy surfaces are ellipsoidal. To begin with we shall calculate the resonance frequency associated with a *single* ellipsoid, which we shall assume to be symmetric about the k_z-axis and to be centered on the point k_{z0}, as shown in Figure 9.18. The magnetic induction $\mathbf{B}_0$ makes an angle θ with the major axis of the ellipsoid, and for convenience we shall assume that the oscillating electric field vector has only a y-component. The choice of this coordinate system is made only for the purposes of the present calculation, and we shall see when we are finished that the only coordinate of physical importance is the angle θ between the field and the longitudinal axis of the ellipsoid.

Under these circumstances, the equation of motion (for a particle of positive charge e) may be written in terms of the tensor (8.8-6) which represents the reciprocal of the effective mass as

$$\frac{d\mathbf{v}}{dt} = \left(\frac{1}{\underline{\mathbf{m}}^*}\right) \cdot \mathbf{F} = \left(\frac{1}{\underline{\mathbf{m}}^*}\right) \cdot \left(e\mathbf{E} + \frac{e}{c}\mathbf{v} \times \mathbf{B}\right). \tag{9.9-16}$$

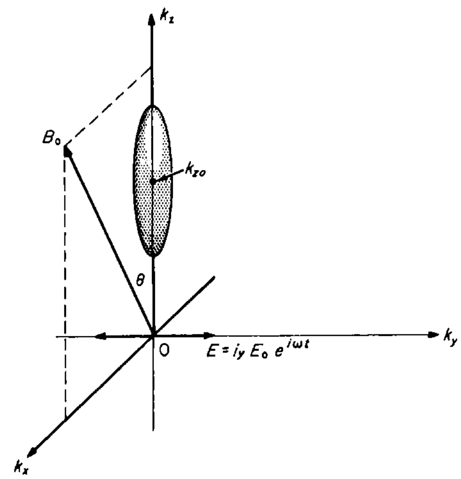


FIGURE 9.18. Vector geometry used for the calculation of Section 9.9.

The relation between ε and $\mathbf{k}$ for the case of the ellipsoidal energy surface shown in Figure 9.18 must, according to (9.9-15) be

$$\varepsilon - \varepsilon_c = \hbar^2\left[\frac{k_x^2 + k_y^2}{2m_\perp^*} + \frac{(k_z - k_{z0})^2}{2m_\parallel^*}\right]. \tag{9.9-17}$$

The tensor components $(1/m^*)_{\alpha\beta}$ may now be found according to (8.8-7). In this case, $\partial^2\varepsilon/\partial k_\alpha \partial k_\beta = 0$ for $\alpha \neq \beta$, so that all the off-diagonal elements of the tensor are zero. The diagonal elements are easily evaluated, the result being

$$(1/m^*)_{xx} = (1/m^*)_{yy} = 1/m_\perp^* \quad \text{and} \quad (1/m^*)_{zz} = 1/m_\parallel^*. \tag{9.9-18}$$

The effective mass tensor then has the form

$$\left(\frac{1}{\underline{\mathbf{m}}^*}\right) = \begin{bmatrix} \frac{1}{m_\perp^*} & 0 & 0 \\ 0 & \frac{1}{m_\perp^*} & 0 \\ 0 & 0 & \frac{1}{m_\parallel^*} \end{bmatrix} \tag{9.9-19}$$

and
$$\left(\frac{1}{\underline{\mathbf{m}}^*}\right)\cdot\mathbf{F} = \begin{bmatrix} \frac{1}{m_\perp^*} & 0 & 0 \\ 0 & \frac{1}{m_\perp^*} & 0 \\ 0' & 0 & \frac{1}{m_\parallel^*} \end{bmatrix} \cdot \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix} = \begin{bmatrix} F_x/m_\perp^* \\ F_y/m_\perp^* \\ F_z/m_\parallel^* \end{bmatrix}. \tag{9.9-20}$$

Inserting the Lorentz force components for F_x, F_y, and F_z and equating to the components of $d\mathbf{v}/dt$ as directed by (9.9-16), the equation of motion can be written

$$\begin{aligned} m_\perp^* \frac{dv_x}{dt} &= \frac{e}{c}(v_yB_z - v_zB_y) + eE_x \\ m_\perp^* \frac{dv_y}{dt} &= \frac{e}{c}(v_zB_x - v_xB_z) + eE_y \\ m_\parallel^* \frac{dv_z}{dt} &= \frac{e}{c}(v_xB_y - v_yB_x) + eE_z. \end{aligned} \tag{9.9-21}$$

Noting that $B_x = B_0 \sin\theta$, $B_z = B_0\cos\theta$, $B_y = 0$, $E_y = E_0e^{i\omega t}$, $E_x = E_z = 0$, these equations may be stated as

$$\begin{aligned} dv_x/dt &= \omega_\perp v_y \cos\theta \\ dv_y/dt &= \omega_\perp v_z \sin\theta - \omega_\perp v_x\cos\theta + \frac{eE_0}{m_\perp^*}e^{i\omega t} \\ dv_z/dt &= -\omega_\parallel v_y \sin\theta \end{aligned} \tag{9.9-22}$$

where
$$\omega_\perp = \frac{eB_0}{m_\perp^* c} \quad \text{and} \quad \omega_\parallel = \frac{eB_0}{m_\parallel^* c}. \tag{9.9-23}$$

As before, one may assume oscillatory solutions of the form $x(t) = x_0e^{i\omega t}$, $y(t) = y_0e^{i\omega t}$, and $z(t) = z_0e^{i\omega t}$, substitute these into the equations of motion (9.9-22) and solve for the amplitudes to obtain

$$\begin{aligned} x_0 &= \frac{eE_0}{m_\perp^*}\frac{i\omega_\perp\cos\theta}{\omega(\omega^2 - \omega_\perp^2\cos^2\theta - \omega_\perp\omega_\parallel\sin^2\theta)} \\ y_0 &= -\frac{eE_0}{m_\perp^*}\frac{1}{\omega^2 - \omega_\perp^2\cos^2\theta - \omega_\perp\omega_\parallel\sin^2\theta} \\ z_0 &= -\frac{eE_0}{m_\perp^*}\frac{i\omega_\parallel\sin\theta}{\omega(\omega^2 - \omega_\perp^2\cos^2\theta - \omega_\perp\omega_\parallel\sin^2\theta)}. \end{aligned} \tag{9.9-24}$$

These amplitudes become very large when

$$\omega = \sqrt{\omega_\perp^2 \cos^2 \theta + \omega_\perp \omega_\parallel \sin^2 \theta}, \tag{9.9-25}$$

and this is accordingly the cyclotron resonance frequency for the particular ellipsoid which we are considering. The resonance frequency depends only upon the magnetic field, the effective masses and the angle θ between the field $\mathbf{B}_0$ and the major axis of the ellipsoid. For ellipsoidal energy surfaces, it is clear that the resonance frequency is a *strong function of the orientation of the sample relative to the magnetic field.* At the resonant frequency the particle orbits can be shown from (9.9-24) to be elliptical in form and to lie in a plane perpendicular to the field $\mathbf{B}_0$. The above results have been derived for positive particles of charge e, but for electrons we need only replace e by $-e$ (in which case we must also replace $\omega_\parallel$ by $-\omega_\parallel$ and $\omega_\perp$ by $-\omega_\perp$). The resonant frequency is obviously unaltered by these substitutions; the only difference is that electrons traverse their orbits in the opposite sense from that found for positive particles.

In the case of silicon, if the field $\mathbf{B}_0$ is oriented along the z-direction, then $\theta = 0$ for the two ellipsoids whose major axes are along the k_z direction and $\theta = 90°$ for the other four ellipsoids. There will then, according to (9.9-25), be two superimposed resonance peaks at $\omega = \omega_\perp$ from the first two ellipsoids and four superimposed resonances at $\omega = \sqrt{\omega_\perp \omega_\parallel}$ from the other four. Assuming that the number of electrons belonging to each ellipsoid is equal, which should be true in equilibrium, the second resonance, involving four ellipsoids should have twice the absorptive strength of the first, which involves only two. If the field $\mathbf{B}_0$ is otherwise oriented, there may be as many as three resonance peaks, corresponding to the three possible angles between $\mathbf{B}_0$ and the ellipsoid axes. This theory of cyclotron resonance with ellipsoidal energy surfaces serves to account for all the experimentally observed resonances which are strongly affected by sample orientation, and allows a determination of both $m_\perp^*$ and $m_\parallel^*$ from the experimentally determined cyclotron resonance frequencies, once a correct model for the orientation of the ellipsoids is proposed.

The cyclotron resonance data for holes in the valence bands of silicon and germanium do not exhibit the features characteristic of the effect with ellipsoidal energy surfaces. Although there are multiple resonance peaks for holes in both materials, the characteristic changes of resonance frequency with sample orientation which are observed for ellipsoidal energy surfaces are either absent or much less pronounced. The picture of the valence band which seems to fit these (and other) data best is that of two separate bands which are degenerate at the energy maximum at $k = 0$, but which have different effective masses. There is also a third "split-off" band with a maximum energy (at $k = 0$) somewhat below the top of the valence band. This state of affairs is shown in Figure 9.19. There are simultaneously present light and heavy holes (belonging to the l and h bands in the figure), and a different cyclotron resonance effect for each type can be observed. Since the energy surfaces associated with these bands are approximately spherical, the orientation dependence of the cyclotron resonance effect is much less pronounced than that which is associated with the ellipsoidal surfaces of the conduction band. Nevertheless, the energy surfaces are only roughly spherical; they are somewhat distorted from a truly spherical form, even for small values of k. This "warping" must be taken into account to explain all the details of the cyclotron resonance data, especially in the case of the heavy hole band, where the departure from sphericity is most pronounced.

The energy difference between the maximum point of the lower-lying "split-off" band and the maximum of the valence band ($\varepsilon_v - \varepsilon_s$ in Figure 9.19) amounts to about 0.28 eV in germanium and 0.035 eV in silicon. This energy difference, in the case of germanium, is sufficiently great that under ordinary conditions the population of holes in the lower band will be negligible. In silicon, however, except at low temperatures, a significant number of holes will be found in this band. The splitting of the bands is caused by the interaction of the electron spins of the valence electrons with

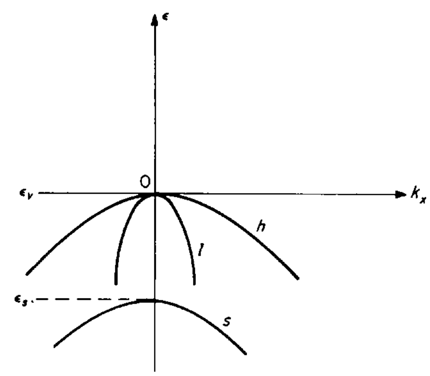


FIGURE 9.19. The three distinct branches of the valence bands of silicon and germanium.

the magnetic moment arising from their orbital motion. If it were not for this *spin-orbit coupling* the energy difference $\varepsilon_v - \varepsilon_s$ would be zero and all three bands would coalesce at the point $k = 0$. The effective masses associated with the three varieties of holes are given in Table 9.3.

TABLE 9.3.

Effective Masses of Holes in Germanium and Silicon

	m_l^*	m_h^*	m_s^*
Germanium	$0.044\ m_0$	$0.28\ m_0$	$0.077\ m_0$
Silicon	$0.16\ m_0$	$0.49\ m_0$	$0.245\ m_0$

The existence of a hole as one particular species is in general limited to a single mean free time. For example, a hole may easily be scattered from the l band to the h-band or vice versa by any of the scattering processes in which it may normally be involved. For this reason it is not possible to detect the existence of the separate varieties of holes in any measurement (such as conductivity or carrier drift) which involves only the average behavior of the carrier over relatively long distances or times, within which many collisions may occur. The results of such a measurement will necessarily only depend upon the properties of the carrier *averaged* over a long time interval during which the identities of holes in the three bands may be frequently interchanged. In the cyclotron resonance experiment, where one measures a resonance which is excited in a time *short* compared to the mean free time ($\omega_0\tau \gg 1$), the presence of the several varieties of holes is easily exhibited. The same remark can be made with reference to the electrons in the various equivalent energy minima of the conduction band; the normal scattering processes may readily transfer electrons from one ellipsoid to another (*intervalley* scattering), so that in a time long compared to the relaxation time, the identity of an electron as belonging to one particular ellipsoid is irretrievably lost.

9·10 DENSITY OF STATES, CONDUCTIVITY AND HALL EFFECT WITH COMPLEX ENERGY SURFACES

In a situation where there are two parabolic bands, degenerate at $k = 0$, with spherical energy surfaces, giving rise to two different species of carriers, the relative populations of the two species and the density of states factors can be easily calculated. We shall discuss the case of a valence band of this sort, in which there are present simultaneously light holes (effective mass m_l^*) and heavy holes (effective mass m_h^*). In this instance we may write the total hole concentration p_0 as the sum of the light hole concentration p_l and the heavy hole concentration p_h, whereby, from (9.3-15) and (9.3-16),

$$p_0 = p_l + p_h = U_{vl}e^{-(\varepsilon_f-\varepsilon_v)/kT} + U_{vh}e^{-(\varepsilon_f-\varepsilon_v)/kT}$$

$$= 2\left(\frac{2\pi kT}{h^2}\right)^{3/2}\left(m_l^{*3/2} + m_h^{*3/2}\right)e^{-(\varepsilon_f-\varepsilon_v)/kT} \tag{9.10-1}$$

where U_{vl} and U_{vh} are factors of the form of (9.3-16) for light and heavy holes. This can be written in the form

$$p_0 = 2\left(\frac{2\pi m_{ds}^* kT}{h^2}\right)^{3/2} e^{-(\varepsilon_f-\varepsilon_v)/kT}, \tag{9.10-2}$$

as if there were a single species of carrier, if we define the *density of states equivalent effective mass* m_{ds}^* as

$$m_{ds}^{*\,3/2} = m_l^{*3/2} + m_h^{*3/2}. \tag{9.10-3}$$

This is an approximation to the actual state of affairs in the valence band of silicon and germanium, although it is a rather rough one, since the effect of the "split-off" band is entirely neglected and since the actual energy surfaces are by no means perfectly spherical. The conductivity follows from (9.10-2) directly, by the use of (9.7-8).

For a set of ellipsoidal energy surfaces such as those discussed in connection with the conduction band of germanium or silicon, one may calculate the density of states factor $g(\varepsilon)$ by an extension of the procedure developed in Section 5.2. The equation (9.9-17) for one of the ellipsoids may be reduced by the transformation

$$p'_x = p_x\sqrt{m_0/m_\perp^*} = \hbar k_x\sqrt{m_0/m_\perp^*}$$

$$p'_y = p_y\sqrt{m_0/m_\perp^*} = \hbar k_y\sqrt{m_0/m_\perp^*}$$

$$p'_z = (p_z - p_{z0})\sqrt{m_0/m_\parallel^*} = \hbar(k_z - k_{z0})\sqrt{m_0/m_\parallel^*} \tag{9.10-4}$$

to that of a sphere in $\mathbf{p}'$-space.

$$\varepsilon - \varepsilon_c = \frac{p_x'^2 + p_y'^2 + p_z'^2}{2m_0} = \frac{p'^2}{2m_0}. \tag{9.10-5}$$

The volume of $\mathbf{p}'$-space in the spherical shell bounded by the radii p' and $p' + dp'$ for a single ellipsoid may now be calculated exactly as in Section 5.2 for spherical energy surfaces, but since there is more than one ellipsoidal constant energy surface, the total volume of $\mathbf{p}'$-space corresponding to energies in the range ε to $\varepsilon + d\varepsilon$ will be obtained by multiplying the volume found for a single surface by the number of equivalent ellipsoids. If this number is ν, then the corresponding volume $dV_{p'}$ of $\mathbf{p}'$-space is

$$dV_{p'} = 4\sqrt{2}\pi\nu m_0^{3/2}\sqrt{\varepsilon - \varepsilon_c}\, d\varepsilon. \tag{9.10-6}$$

But, from (9.10-4) the volume elements $dV_p = dp_x dp_y dp_z$ and $dV_{p'} = dp_{x'} dp_{y'} dp_{z'}$ are related by

$$dV_{p'} = \frac{m_0^{3/2}}{(m_\perp^{*2} m_\parallel^*)^{1/2}}\, dV_p, \tag{9.10-7}$$

whereby the volume dV_p of $\mathbf{p}$-space within the energy range ε to $\varepsilon + d\varepsilon$ becomes

$$dV_p = 4\sqrt{2}\pi\nu(m_\perp^{*2} m_\parallel^*)^{1/2}\sqrt{\varepsilon - \varepsilon_c}\, d\varepsilon. \tag{9.10-8}$$

Dividing this by the volume $h^3/2$ assigned to a single quantum state gives the number of quantum states in this energy range which, by definition, is $g(\varepsilon)\, d\varepsilon$, whereby

$$g(\varepsilon)\, d\varepsilon = \frac{8\sqrt{2}\pi\nu(m_\perp^{*2} m_\parallel^*)^{1/2}}{h^3}\sqrt{\varepsilon - \varepsilon_c}\, d\varepsilon. \tag{9.10-9}$$

If, once more, we define a *density of states equivalent effective mass* m_{ds}^* as

$$m_{ds}^{*3/2} = (m_\perp^{*2} m_\parallel^*)^{1/2}, \tag{9.10-10}$$

Equation (9.10-9) can be written (except for the factor ν) in the same form (5.2-22) as the usual density of states expression for the case where the energy surfaces are spherical. By a straightforward application of the procedures by which (9.3-6) and (9.3-7) were obtained, it is easily seen that we must obtain

$$n_0 = 2\nu\left(\frac{2\pi m_{ds}^* kT}{h^2}\right)^{3/2} e^{-(\varepsilon_c - \varepsilon_f)/kT} \tag{9.10-11}$$

with m_{ds}^* as given by (9.10-10).

In calculating the conductivity of a semiconductor in which the constant energy surfaces are ellipsoidal, we must first calculate the current or conductivity contribution from a single ellipsoid and then sum over all the ellipsoids to obtain the total conductivity. Since the effective mass associated with each individual ellipsoid is a tensor quantity, so also is the conductivity. In general, the equations relating the current density and electric field may be written as tensor relations of the form

$$\mathbf{I} = \underline{\boldsymbol{\sigma}} \cdot \mathbf{E} \tag{9.10-12}$$

and

$$\mathbf{E} = \underline{\boldsymbol{\rho}} \cdot \mathbf{I} \tag{9.10-13}$$

where $\underline{\boldsymbol{\sigma}}$ is the *conductivity tensor* and $\underline{\boldsymbol{\rho}}$ is a resistivity tensor. Since, clearly, from (9.10-12) and (9.10-13),

$$\underline{\boldsymbol{\rho}} \cdot \mathbf{I} = \underline{\boldsymbol{\rho}} \cdot \underline{\boldsymbol{\sigma}} \cdot \mathbf{E} = \mathbf{E}, \tag{9.10-14}$$

we must have

$$\underline{\boldsymbol{\rho}} \cdot \underline{\boldsymbol{\sigma}} = \underline{\mathbf{1}}, \tag{9.10-15}$$

where $\underline{\mathbf{1}}$ is the unit tensor, whose elements are $\delta_{\alpha\beta}$. The resistivity tensor is thus the inverse of the conductivity tensor, that is,

$$\underline{\boldsymbol{\rho}} = \underline{\boldsymbol{\sigma}}^{-1}. \tag{9.10-16}$$

The elements of the conductivity tensor $\underline{\boldsymbol{\sigma}}^{(i)}$ related to the ith energy ellipsoid may in analogy with (7.3-15) be written

$$\sigma_{\alpha\beta}^{(i)} = n^{(i)} e^2 \bar{\tau} \left(\frac{1}{m^*}\right)_{\alpha\beta} \tag{9.10-17}$$

where $n^{(i)}$ is the density of charge carriers associated with that ellipsoid. The total conductivity may then be expressed as

$$\underline{\boldsymbol{\sigma}} = \sum_i \underline{\boldsymbol{\sigma}}^{(i)}, \tag{9.10-18}$$

the sum being taken over all ellipsoids. Using (8.8-7) to express the tensor components, (9.10-17) can be written as

$$\underline{\boldsymbol{\sigma}}^{(i)} = \frac{n^{(i)} e^2 \bar{\tau}}{\hbar^2} \begin{bmatrix} \dfrac{\partial^2 \varepsilon^{(i)}}{\partial k_x^2} & \dfrac{\partial^2 \varepsilon^{(i)}}{\partial k_x \, \partial k_y} & \dfrac{\partial^2 \varepsilon^{(i)}}{\partial k_x \, \partial k_z} \\ \dfrac{\partial^2 \varepsilon^{(i)}}{\partial k_y \, \partial k_x} & \dfrac{\partial^2 \varepsilon^{(i)}}{\partial k_y^2} & \dfrac{\partial^2 \varepsilon^{(i)}}{\partial k_y \, \partial k_z} \\ \dfrac{\partial^2 \varepsilon^{(i)}}{\partial k_z \, \partial k_x} & \dfrac{\partial^2 \varepsilon^{(i)}}{\partial k_z \, \partial k_y} & \dfrac{\partial^2 \varepsilon^{(i)}}{\partial k_z^2} \end{bmatrix} \tag{9.10-19}$$

where $\varepsilon^{(i)}(\mathbf{k})$ is the energy expressed as a function of $\mathbf{k}$ for the ith ellipsoid. This treatment assumes, of course, that the relaxation time τ is the same for all possible directions in $\mathbf{k}$-space, which may not be true in the most general case.

In the case of silicon, the major axes of the ellipsoids lie along the coordinate axes of **k**-space; for the two ellipsoids whose major axes lie along the k_z-axis, in the [001] and [00$\bar{1}$] directions the function $\varepsilon^{(001)}(\mathbf{k})$ is given by (9.9-17), leading to tensor components (9.9-18) for the effective mass tensor. The conductivity tensor for these ellipsoids then becomes

$$\underline{\boldsymbol{\sigma}}^{(001)} = \underline{\boldsymbol{\sigma}}^{(00\bar{1})} = n^{(001)}e^2\bar{\tau}\begin{bmatrix} \frac{1}{m_\perp^*} & 0 & 0 \\ 0 & \frac{1}{m_\perp^*} & 0 \\ 0 & 0 & \frac{1}{m_\parallel^*} \end{bmatrix} \tag{9.10-20}$$

The equations for the other ellipsoids may be written and the tensor components obtained in the same way. For the two ellipsoids whose major axes lie along the k_x-axis, then

$$\underline{\boldsymbol{\sigma}}^{(100)} = \underline{\boldsymbol{\sigma}}^{(\bar{1}00)} = n^{(100)}e^2\bar{\tau}\begin{bmatrix} \frac{1}{m_\parallel^*} & 0 & 0 \\ 0 & \frac{1}{m_\perp^*} & 0 \\ 0 & 0 & \frac{1}{m_\perp^*} \end{bmatrix}, \tag{9.10-21}$$

while for the two ellipsoids whose major axes lie along the k_y-axis,

$$\underline{\boldsymbol{\sigma}}^{(010)} = \underline{\boldsymbol{\sigma}}^{(0\bar{1}0)} = n^{(010)}e^2\bar{\tau}\begin{bmatrix} \frac{1}{m_\perp^*} & 0 & 0 \\ 0 & \frac{1}{m_\parallel^*} & 0 \\ 0 & 0 & \frac{1}{m_\perp^*} \end{bmatrix} \tag{9.10-22}$$

Since all six ellipsoids are equivalent energy minima, $n^{(100)} = n^{(\bar{1}00)} = n^{(010)} = n^{(0\bar{1}0)} = n^{(001)} = n^{(00\bar{1})} = n_0/6$. Performing the summation indicated in(9.10-18)and utilizing this fact, one may obtain

$$\underline{\boldsymbol{\sigma}}_0 = n_0e^2\bar{\tau}\begin{bmatrix} \frac{1}{3}\left(\frac{2}{m_\perp^*} + \frac{1}{m_\parallel^*}\right) & 0 & 0 \\ 0 & \frac{1}{3}\left(\frac{2}{m_\perp^*} + \frac{1}{m_\parallel^*}\right) & 0 \\ 0 & 0 & \frac{1}{3}\left(\frac{2}{m_\perp^*} + \frac{1}{m_\parallel^*}\right) \end{bmatrix}$$

$$= n_0e^2\bar{\tau}\,\frac{1}{3}\left(\frac{2}{m_\perp^*} + \frac{1}{m_\parallel^*}\right)\underline{\mathbf{1}} \tag{9.10-23}$$

for the total conductivity. Since $\underline{\sigma}$ is a scalar multiple of the unit tensor $\underline{1}$, the *total* conductivity is *isotropic* (albeit the conductivity associated with a single ellipsoid is not). The scalar magnitude of the conductivity can be represented as

$$\sigma_0 = \frac{n_0 e^2 \bar{\tau}}{m_c^*} \tag{9.10-24}$$

where m_c^* is a *conductivity effective mass* defined by

$$\frac{1}{m_c^*} = \frac{1}{3}\left(\frac{2}{m_\perp^*} + \frac{1}{m_\parallel^*}\right). \tag{9.10-25}$$

It can be shown that exactly the same result will be obtained for *any* set of ellipsoidal energy surfaces whatever, so long as the configuration is invariant under all of the symmetry operations under which a cubic crystal is invariant. In fact, it can be proved that the conductivity of all cubic crystals must be isotropic, whatever specific form the surfaces of constant energy take. In particular, Equations (9.10-23) through (9.10-25) are valid *also* for the conduction electrons in germanium, where the constant energy ellipsoids lie along the $\langle 111 \rangle$ directions. It is instructive to work out the conductivity tensor for germanium; this problem is assigned as an exercise for the reader.

The resistivity tensor $\underline{\rho}$ is a matrix of components inverse to the conductivity matrix. The elements of the matrix which is the inverse of a matrix $\underline{A}$ with elements $a_{\alpha\beta}$ are given by the formula

$$a_{\alpha\beta}^{-1} = \frac{\operatorname{cof}(a_{\beta\alpha})}{\Delta(a)}, \tag{9.10-26}$$

where $\operatorname{cof}(a_{\beta\alpha})$ is the cofactor[10] of the element $a_{\beta\alpha}$ and $\Delta(a)$ is the determinant of the coefficients of $\underline{A}$. The elements of $\underline{\rho}$ are thus readily seen to be

$$\underline{\rho} = \underline{\sigma^{-1}} = \frac{1}{\Delta(\sigma)}\begin{bmatrix} \sigma_{yy}\sigma_{zz} - \sigma_{yz}^2 & \sigma_{xz}\sigma_{yz} - \sigma_{xy}\sigma_{zz} & \sigma_{xy}\sigma_{yz} - \sigma_{xz}\sigma_{yy} \\ \sigma_{yz}\sigma_{xz} - \sigma_{xy}\sigma_{zz} & \sigma_{xx}\sigma_{zz} - \sigma_{xz}^2 & \sigma_{xz}\sigma_{xy} - \sigma_{xx}\sigma_{yz} \\ \sigma_{xy}\sigma_{yz} - \sigma_{xz}\sigma_{yy} & \sigma_{xz}\sigma_{xy} - \sigma_{xx}\sigma_{yz} & \sigma_{xx}\sigma_{yy} - \sigma_{xy}^2 \end{bmatrix} \tag{9.10-27}$$

where
$$\Delta(\sigma) = \sigma_{xx}(\sigma_{yy}\sigma_{zz} - \sigma_{yz}^2) - \sigma_{xy}(\sigma_{xy}\sigma_{zz} - \sigma_{yz}\sigma_{xz}) + \sigma_{xz}(\sigma_{xy}\sigma_{yz} - \sigma_{yy}\sigma_{xz}). \tag{9.10-28}$$

It is readily seen that if $\underline{\sigma}$ has the form (9.10-23), the resistivity tensor is isotropic and can be represented by a scalar quantity whose magnitude is simply the reciprocal of the conductivity (9.10-24).

We shall illustrate the computation of the Hall effect in a material with ellipsoidal energy surfaces by considering in detail the Hall effect in a material where the major axes of the ellipsoids extend along the k_x-, k_y- and k_z-axes (as, for example, in silicon),

[10] See, for example, L. A. Pipes, *Applied Mathematics for Engineers and Physicists*. New York: McGraw-Hill (1946), p. 71.